

COMBINATORICS

EXAM 1

ROB TOMPKINS

1. Answer the following using simple counting reasoning

- (a) How many different 7-digit numbers (non-negative integers) are there? (Leading zeros are not allowed).
- (b) How many even 7-digit numbers are there?
- (c) How many 7-digit numbers are there with exactly one 7?
- (d) How many 7-digit numbers read the same when you read them forward or backward, i.e. are palindromic, e.g. 6548456 is one such integer.
- (e) How many 7-digit numbers are there that are divisible by 4?
- (f) How many times is the digit 7 written when listing the numbers between 1 and 1,000,000?

Solution.

□

2. How many 7-letter words can be constructed by using the 26 letters of the alphabet if each word contains 3, or 4 vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution.

□

3. Let n and k be positive integers. Give a combinatorial proof that

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

Proof.

□

4. In how many ways can we select a set of m marbles from a collection of $2m$ marbles, where m of them are identical and the remaining m are all distinct

Solution.

□